

Advantages and Disadvantages of Optoelectronic Sensing

Advantages. High sensitivity, chemically inert, small and lightweight, suitable for remote sensing, immune to electromagnetic interference, wide dynamic range, capable of monitoring a wide range of chemical and physical parameters, and reliable operation

Disadvantages. Susceptible to interference from environmental effects, can be costly and susceptible to physical damage

emitting and light-detecting devices. Optoelectronic devices and components operate on both light and electrical currents.

Optoelectronics-based technology refers to electronic device applications for the sourcing, detection and control of light. It encompasses design, manufacture and study of electronic hardware devices that, as a result, convert electricity into photon signals for purposes of sensing, medical diagnostics, telecommunications and general science, among others.

Optoelectronics deals with radiation, its detection, creation and manipulation for various purposes. This includes x-rays, gamma rays, infrared (IR), ultraviolet (UV) and visible light.

Optoelectronic devices are basically transducers that convert electrical or optical forms of energy into one another. These can either be electrical-to-optical or optical-to-electrical. Thus, the devices act as detectors of light and transform detected light signals into equivalent electrical signals for computer processing.

Quantum mechanical effects of light on semiconducting materials used in electronic devices based on optoelectronics are as follows:

Photovoltaic or photoelectric. This involves direct conversion of light into electricity using solar cells.

Photoconductivity. This is an electrical phenomenon wherein a material becomes more conductive to electricity through the absorption of electromagnetic radiation such as IR, UV and visible light.

Stimulated emission. It is a process where a light photon interacts with an excited molecule, which causes it to drop to a lower energy level, resulting in the emission/liberation of an identical photon. This photon is transferred to the electro-

magnetic field used in laser diodes and quantum cascade lasers.

Radiative recombination. Here, electrons are transitioned from valence to conducting band in semiconductors, resulting in a carrier generation and recombination effect, which produces light.

Technological developments

Optoelectronic devices find applications in fields like military services, telecommunications, automatic access control systems, medical equipment and detecting food adulteration. The field covers a wide range of devices including LEDs and elements, image pickup devices, information displays, optical communication systems, optical storages and remote sensing systems.

Some optoelectronic devices that are currently making headlines are telecommunication lasers, blue lasers, optical fibres, LED traffic lights, photo diodes and solar cells. Some latest technological developments taking place in the field of optoelectronic sensing are summarised below.

1. Suitable integration of advanced IR spectroscopy with chemometrics provides a powerful tool for adulteration detection.

2. An electronic tongue is an easy and economic tool for the detection of adulterants.

3. Biosensors with the ability to furnish real-time signals have been developed for the detection of urea in milk.

4. An embedded optoelectronic sensing system consisting of AVR microcontroller integrated with optical sensor, LCD and keypad, making use of the change in refractive index parameter, can detect milk adulteration with an accuracy of about 95 per cent. This is an increase in accuracy of about 200 per cent over traditional methods such as spectroscopy.

5. Fibre-optics-based optoelectronic sensing systems can be used for adulteration detection.

6. Spectral changes in the grating response due to variations in adulteration levels can be used as investigative tools. The fundamental principle of detection is the sensitive dependence of resonance peaks on the changes in the refractive index of the medium.

7. Surface plasmon resonance (SPR) can be employed as an alternative method to detect adulteration. SPR sensors offer the capability of detection of very low concentrations of adulterated chemical and biological agents in real time by monitoring.

Adulterants change the refractive index of food, resulting in corresponding phase changes. These changes can be detected at the output and related to the concentration of chemical species present at the point.

Shift of resonance angle as a function of adulterants concentration can be plotted with linear regression. The outcome has disclosed a real-time, rapid and non-destructive sensor to be promoted as well-developed.

Conclusion

Nowadays, adulteration is being done in sophisticated ways that demand cutting-edge research for the detection of adulterants. Conventional chromatography-based methods for the detection of chemicals are based on human observation and slow for real-time monitoring. Keeping in mind the limitations of traditional methods of different adulterants detection, electronic sensing is the technology to count on.

Further, optoelectronic devices are more sensitive, reliable and suitable to detect different types of adulterants. With growing concern about adulteration and strong footing in science and technology of optoelectronic devices, it is expected that, soon people will be equipped with cost-effective, environment-friendly, user-friendly, reliable, easy-to-operate optoelectronic sensing devices to check for any type and level of food adulteration. **EFY**